Language Models are Few-Shot Learners (Brown et al. 2020)

Scaling autoregressive Transformer language models to 175 billion parameters unlocks in-context few-shot learning capabilities across diverse tasks without task-specific gradient fine-tuning.

GRADE: HIGH | STATUS: STABLE | DOMAIN: TECHNICAL, ORGANIZATIONAL, PHILOSOPHICAL | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Scaling autoregressive Transformer models by orders of magnitude (to 175B parameters in GPT-3) enables few-shot in-context learning where models perform novel linguistic and cognitive tasks guided solely by text prompts in their context window without fine-tuning weights.

EVIDENCE AND CAUSAL ASSESSMENT

Controlled evaluations comparing zero-shot, one-shot, and few-shot conditions across 40+ datasets isolate prompt demonstration length and parameter scale as the causal determinants of task accuracy.

EFFECT SIZE AND PERSISTENCE

On SuperGLUE, GPT-3 175B achieved 71.8 in few-shot mode, matching fine-tuned BERT models and demonstrating strong qualitative jumps on translation, arithmetic, and cloze tasks.

BIAS, HETEROGENEITY, AND CONFOUNDING

Classified as robust (<20%). In-context learning performance remains brittle to prompt formatting, token ordering, and demonstration selection, and models can retrieve confident hallucinations when context is ambiguous.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

This does not mean that the model has acquired new permanent memories; once the context window is cleared, all few-shot demonstrations vanish, leaving the underlying weights unchanged.

WHAT ACTION IS REASONABLY JUSTIFIED

Use prompt engineering and few-shot in-context examples for agile task prototyping while relying on external vector retrieval (RAG) for verifiable domain knowledge.

VERIFIED SOURCES & DOI IDENTIFIERS

- Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., ... & Amodei, D. (2020). Language Models are Few-Shot Learners. NeurIPS 2020. DOI: 10.48550/arXiv.2005.14165